CHAPTER 10

Mensuration

It is one of the easiest chapters, which contributes almost 6-8% problems in Quantitative Aptitude Section of CAT. Besides there are several other aptitude tests which include plethora of questions from this topic itself.

Therefore it is advised that those students who are not so good in other sections such as algebra or sort of logical questions they must emphasise on this chapter. Even the questions asked from this chapter are not as much complex as they are in Geometry.

10.1 Mensuration

Definition : Mensuration is a science of measurement of the lengths of lines, areas of surfaces and volumes of solids.

Planes : Planes are two dimensional i.e. these two dimensions are namely length and breadth. These occupy surface.

Solids : Solids are three dimensional, namely length, breadth and height. These occupy space.

Conversion of Important Units

1 km = 10 hectometre = 100 decametre
= 1000 metre = 10,000 decimetre
= 1,00,000 centimetre = 10,00,000 millimetre

1 hectare = 10,000 square metre

1 are = 100 square metre

1 square hectometre = 100 square decametre

1 square decametre = 100 square metre

1 square metre = 100 square decimetre

1 square decimetre = 100 square centimetre

1 square centimetre = 100 square millimetre

$\sqrt{2} = 1.414, \quad \sqrt{3} = 1.732, \quad \sqrt{5} = 2.236, \quad \sqrt{6} = 2.45$

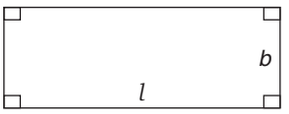
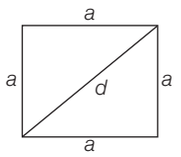
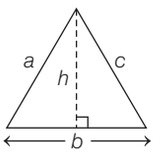
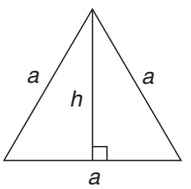
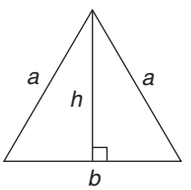
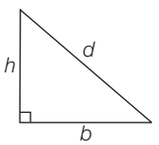
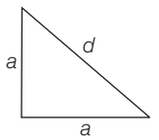
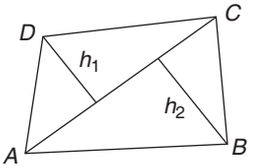
Weight = Volume $\times$ density

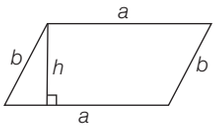
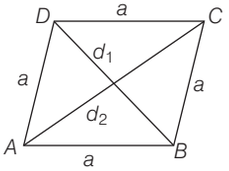
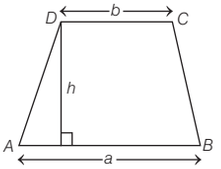
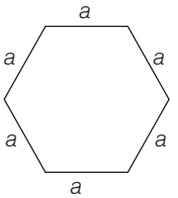
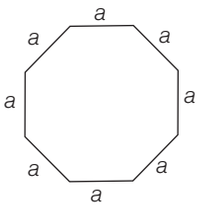
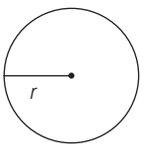
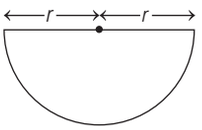
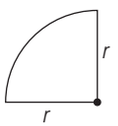
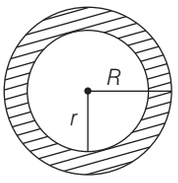
Chapter Checklist

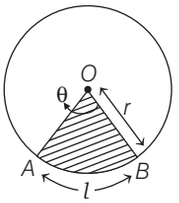
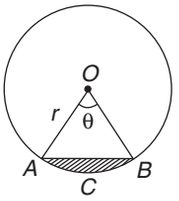
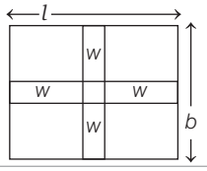
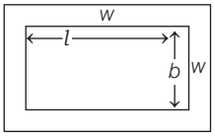
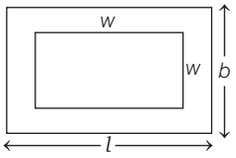
- *Mensuration*
- *2-D Figures (Planes)*
- *Rectangles and Squares*
- *Triangles*
- *Parallelogram, Rhombus and Trapezium*
- *Circles*
- *3-D Figures (Solids)*
- *Cuboid and Cube*
- *Cylinder and Cone*
- *Sphere, Prism and Pyramid*
- ***CAT Test***

10.2 2-D Figures

Table 2-D Figures (Plane figures)

Name	Figure	Nomenclature	Area	Perimeter
Rectangle		$l \rightarrow$ length $b \rightarrow$ breadth	$l \times b = lb$	$2l + 2b = 2(l + b)$
Square		$a \rightarrow$ side $d \rightarrow$ diagonal $d = a\sqrt{2}$	(i) $a \times a = a^2$ (ii) $\frac{d^2}{2}$	$a + a + a + a = 4a$
Triangle (Scalene)		a, b and c are three sides of triangle and s the semiperimeter, where $s = \left(\frac{a+b+c}{2}\right)$ b is the base and h is the altitude of triangle	(i) $\frac{1}{2} \times b \times h$ (ii) $\sqrt{s(s-a)(s-b)(s-c)}$ (Hero's formula)	$a + b + c = 2s$
Equilateral triangle		$a \rightarrow$ side $h \rightarrow$ height or altitude $h = \frac{\sqrt{3}}{2}a$	(i) $\frac{1}{2} \times a \times h$ (ii) $\frac{\sqrt{3}}{4}a^2$	$3a$
Isosceles triangle		$a \rightarrow$ equal sides $b \rightarrow$ base $h \rightarrow$ height or altitude $h = \frac{\sqrt{4a^2 - b^2}}{2}$	(i) $\frac{1}{2} \times b \times h$ (ii) $\frac{1}{4} \times b \times \sqrt{4a^2 - b^2}$	$2a + b$
Right angled triangle		$b \rightarrow$ base $h \rightarrow$ altitude/height $d \rightarrow$ diagonal $d = \sqrt{b^2 + h^2}$	$\frac{1}{2} \times b \times h$	$b + h + d$
Isosceles right angled triangle		$a \rightarrow$ equal sides $d \rightarrow$ diagonal $d = a\sqrt{2}$	$\frac{1}{2}a^2$	$2a + d$
Quadrilateral		AC is the diagonal and h_1, h_2 are the altitudes on AC, from the vertices D and B, respectively.	$\frac{1}{2} \times AC \times (h_1 + h_2)$	$AB + BC + CD + AD$

Name	Figure	Nomenclature	Area	Perimeter
Parallelogram		a and b are sides adjacent to each other. $h \rightarrow$ distance between the parallel sides	$a \times h$	$2(a + b)$
Rhombus		$a \rightarrow$ each equal side of rhombus d_1 and d_2 are the diagonals $d_1 \rightarrow BD$ $d_2 \rightarrow AC$	$\frac{1}{2} \times d_1 \times d_2$	$4a$
Trapezium		a and b are parallel sides to each other and h is the perpendicular distance between parallel sides	$\left(\frac{a+b}{2}\right) \times h$	$AB + BC + CD + AD$
Regular hexagon		$a \rightarrow$ each of the equal side	$\frac{3\sqrt{3}}{2} a^2$	$6a$
Regular octagon		$a \rightarrow$ each of equal side	$2a^2 (1 + \sqrt{2})$	$8a$
Circle		$r \rightarrow$ radius of the circle $\pi = \frac{22}{7} = 3.1416$ (approx)	πr^2	$2\pi r$ (called as circumference)
Semicircle		$r \rightarrow$ radius of the circle	$\frac{1}{2} \pi r^2$	$\pi r + 2r$
Quadrant		$r \rightarrow$ radius	$\frac{1}{4} \pi r^2$	$\frac{1}{2} \pi r + 2r$
Ring or circular path (shaded region)		$R \rightarrow$ outer radius $r \rightarrow$ inner radius	$\pi (R^2 - r^2)$	(outer) $\rightarrow 2\pi R$ (inner) $\rightarrow 2\pi r$

Name	Figure	Nomenclature	Area	Perimeter
Sector of a circle		$O \rightarrow$ centre of the circle $r \rightarrow$ radius $l \rightarrow$ length of the arc $\theta \rightarrow$ angle of the sector $l = 2\pi r\left(\frac{\theta}{360^\circ}\right)$	(i) $\pi r^2\left(\frac{\theta}{360^\circ}\right)$ (ii) $\frac{1}{2} r \times l$	$l + 2r$
Segment of a circle		$\theta \rightarrow$ angle of the sector $r \rightarrow$ radius $AB \rightarrow$ chord $ACB \rightarrow$ arc of the circle	Area of segment ACB (minor segment) $= r^2\left(\frac{\pi\theta}{360^\circ} - \frac{\sin\theta}{2}\right)$	$2r\left[\frac{\pi\theta}{360^\circ} + \sin\left(\frac{\theta}{2}\right)\right]$
Pathways running across the middle of a rectangle		$l \rightarrow$ length $b \rightarrow$ breadth $w \rightarrow$ width of the path (road)	$(l + b - w)\, w$	$2(l + b) - 4w$ $= 2[l + b - 2w]$
Outer pathways		$l \rightarrow$ length $b \rightarrow$ breadth $w \rightarrow$ widthness of the path	$(l + b + 2w)\, 2w$	(inner) $\rightarrow 2(l + b)$ (outer) $\rightarrow 2(l + b + 4w)$
Inner path		$l \rightarrow$ length $b \rightarrow$ breadth $w \rightarrow$ widthness of the path	$(l + b - 2w)\, 2w$	(outer) $\rightarrow 2(l + b)$ (inner) $\rightarrow 2(l + b - 4w)$

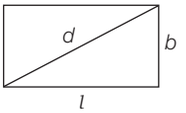
10.3 Rectangles and Squares

1. Area of a rectangle = length × breadth = $l \times b$
2. Area of a square = $(\text{side})^2 = \frac{1}{2}(\text{diagonal})^2$
$$= \frac{1}{2} d^2 = a^2$$
3. Diagonal of a rectangle
$$= \sqrt{(\text{length})^2 + (\text{breadth})^2} = \sqrt{l^2 + b^2}$$
4. Diagonal of a square $= \sqrt{\text{side}^2 + \text{side}^2} = \text{side}\ \sqrt{2} = a\sqrt{2}$
5. Perimeter of a rectangle $= 2$ (length + breadth) $= 2(l + b)$
6. Perimeter of a square $= 4 \times \text{side} = 4a$
7. Area of four walls of a room $= 2(l + b) \times h$

Exp. 1) The length and breadth of a rectangular room are 15 m and 8 m respectively :
(a) Find the perimeter of room.
(b) Find the area of the floor of room.
(c) Find the maximum possible length of the rod that can be put on the floor.

Solution
(a) Perimeter = 2(15 + 8) = 46 m
(b) Area of floor = 15 × 8 = 120 m^2
(c) Length of diagonal = $\sqrt{(15)^2 + (8)^2} = 17$ m

NOTE The maximum possible length of any rod that can be placed on a rectangular floor is equal to the diagonal of the floor of the room.



Here $d > l$

Exp. 2) The side of a square shaped garden is 20 m. Find the :
(a) area of the garden
(b) perimeter (or boundary) of the garden
(c) maximum possible distance between any two corners of the garden.

Solution (a) Area = $(\text{side})^2 = (20)^2 = 400\ m^2$ (square metre)
(b) Perimeter = 4 × side = 4 × 20 = 80 m
(c) Diagonal = side $\sqrt{2} = 20 \times \sqrt{2} = 20 \times 1.414 = 28.28$ m

***Exp.* 3)** One side of a rectangular lawn is 12 m and its diagonal is 13 m. Find the area of the field.

Solution $d = \sqrt{l^2 + b^2} \Rightarrow 13 = \sqrt{12^2 + b^2} \Rightarrow b = 5\text{ m}$

$\therefore$ Area $= l \times b = 12 \times 5 = 60\text{ m}^2$

***Exp.* 4)** The length of a rectangle is 1 cm more than its breadth. The diagonal is 29 cm. Find the area of the rectangle.

(a) 481 cm^2 (b) 841 cm^2 (c) 420 cm^2 (d) 870 m^2

Solution $l = (b + 1)$

and $d = \sqrt{l^2 + b^2} = \sqrt{(b+1)^2 + b^2} = 29$

$\Rightarrow$ $b^2 + b = 420 \Rightarrow b = 20$

$\therefore$ $l = 21$

$\therefore$ Area $= l \times b = 420\text{ cm}^2$

***Exp.* 5)** The length of a wall is 5/4 times of its height. If the area of the wall be 180 m^2. What is the sum of the length and height of the wall?

Solution Let the length be $5x$ and height be $4x$.

Then, $l \times h = 180 = 5x \times 4x = 20x^2 \Rightarrow x = 3$

$\therefore$ $l + h = 15 + 12 = 27\text{ m}$

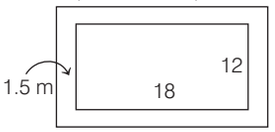
Alternatively $1.25\ h^2 = 180 \Rightarrow h = 12\text{ m}$

$\therefore$ $l = 15\text{ m}; l + h = 27\text{ m}$

***Exp.* 6)** A rectangular grassy lawn is 18 m by 12 m. It has a gravel path 1.5 m wide all around it on the outside. What is the area of the path.

Solution Area of path (outside the lawn) $= (l + b + 2w)\ 2w$

$= (18 + 12 + 3)\ 3 = 99\text{ m}^2$



Alternatively Area of lawn $= 18 \times 12 = 216\text{ m}^2$

Total area (lawn + path) $= (18 + 3) \times (12 + 3)$

$= 21 \times 15 = 315\text{ m}^2$

$\therefore$ Area of path (only) $= 315 - 216 = 99\text{ m}^2$

***Exp.* 7)** Find the cost of carpeting a room 17 m long and 9 m wide with a carpet 60 cm broad at 40 paise per metre.

Solution Area of carpet = Area of room

$l \times 0.6 = 17 \times 9$ (60 cm = 0.6 m)

$\Rightarrow$ $l = 255$ m

$\therefore$ Cost of carpeting the floor = rate $\times$ length of carpet

$= 0.4 \times 255 = ₹\ 102.00$

***Exp.* 8)** The dimensions of a lawn are in the ratio 4 : 1 and its area is 1/4 hectares. What is the length of the lawn?

Solution $l \times b = 4x \times x = \frac{1}{4} \times 10000$

$\Rightarrow$ $x = 25$

$\Rightarrow$ length $= 4x = 100$ m

***Exp.* 9)** A room is 16 m long, 7 m broad and 8 m high. Find the cost of white washing the four walls of room at ₹ 7.5 per m^2, white washing is not to be done on the doors and windows, which occupy 65 m^2.

Solution Area of 4 walls of a room $= 2(16 + 7) \times 8 = 368\text{ m}^2$

Net area of 4 walls $= 368 - 65 = 303\text{ m}^2$

$\therefore$ Cost of white washing $= 303 \times 7.5 = ₹\ 2272.5$

Exp. 10) The ratio between the sides of a room is 5 : 3. The cost of white washing the ceiling of the room at 50 Paise per square m is ₹ 270 and the cost of papering the walls at 10 P per square metre is ₹ 48. The height of the room is :

(a) 6 m (b) 8 m

(c) 5 m (d) 10 m

Solution Area of ceiling $= \frac{\text{Total cost}}{\text{Cost of 1 sq unit}} = \frac{270}{0.5} = 540\text{ sq m}$

Now since $l : b = 5x : 3x$

$\Rightarrow$ $l \times b = 15x^2 = 540\text{ m}^2$

$\Rightarrow$ $l = 30$ and $b = 18$ m

Now, Area of the 4 walls

$= \frac{\text{Total cost}}{\text{Cost of 1 sq unit}} = \frac{48}{0.1} = 480\text{ m}^2$

$\therefore$ Height $= \frac{480}{2(30 + 18)} = 5$ m

Introductory Exercise 10.1

1. A rectangular field has its length and breadth in the ratio of 16 : 9. If its perimeter is 750 cm. What is its area?
(a) 7500 cm^2 (b) 32400 cm^2
(c) 14400 cm^2 (d) 14000 cm^2

2. A rectangular field costs ₹ 110 for levelling at 50 paise per square metre. If the ratio of length : breadth is 11 : 5. Find the length of the field.
(a) 16 m (b) 21 m
(c) 22 m (d) none of these

3. A room is half as long again as it is wide. The cost of carpeting it at 62 paise per square metre is ₹ 2916.48. Find the cost of white washing the ceiling at 30 paise per metre.
(a) ₹ 2211.5 (b) ₹ 1114.2
(c) ₹ 1411.2 (d) can't be determined

4. The length of a rectangular plot of ground is four times its breadth and its area is 4 hectares. How long will it take to a dog to walk round it at the rate of 3 km/h?
(a) 12 min (b) 20 min (c) 21 min (d) 18.5 min

5. Find the length of the wire required to fence a square field 6 times having its area 5 hectares and 76 ares.
(a) 5760 m (b) 6760 m (c) 52500 m (d) 11760 m

6. A room is 19 m long and 3.50 m broad. What will be the cost of covering its floor with a carpet of 70 cm wide at 95 paise per metre?
(a) ₹ 90.25 (b) ₹ 99.25
(c) ₹ 90.75 (d) none of these

7. Find the cost of paving a courtyard 316.8 m × 65 m with stones measuring 1.3 m × 1.1 m at ₹ 0.5 per stone.
(a) ₹ 1440 (b) ₹ 7200
(c) ₹ 72,000 (d) none of these

8. A rectangular garden 63 m long and 54 m broad has a path 3 m wide inside it. Find the cost of paving the path at ₹ 37/2 per square metre.
(a) ₹ 12321 (b) ₹ 11100
(c) ₹ 74000 (d) none of these

9. If the length of a ractangular field is doubled and its breadth is halved (i.e. reduced by 50%). What is percentage change in its area?
(a) 0% (b) 10% (c) 25% (d) 33.33%

10. A path of uniform width runs all around the inside of rectangular field 116 m by 68 m and occupies 720 sq m. Find the width of the path.
(a) 1 m (b) 1.5 m (c) 2 m (d) 4 m

11. A drawing room is 7.5 m long 6.5 m broad and 6 m high. Find the length of paper 2.5 dm wide to cover its walls allowing 8 sq m for doors.
(a) 368 m (b) 640 m (c) 625 m (d) 888 m

12. A square field of 2 sq km is to be divided into two equal parts by a wall which coincides with a diagonal. Find the length of the wall.
(a) $\sqrt{2}$ km (b) 1 km (c) 4.2 km (d) 2 km

13. There are two square fields. Of the two square fields one contains 1 hectare area while the other is broader by 11 per cent. Find the difference in area expressed in sq m.
(a) 2321 sq m (b) 1210 sq m
(c) 2121 sq m (d) 7700 sq m

14. The expenses of carpeting a half of the floor were ₹ 759, but if the length had been 6 m less than it was, the expenses would have been ₹ 561. What is the length?
(a) 21 m (b) 23 m (c) 45 m (d) 27 m

15. If a roll of paper 1 km long has area 1/25 hectare, how wide is the paper?
(a) 4 m (b) 40 cm (c) 40 dm (d) 25 cm

16. The diagonal and one side of a rectangular plot are 289 m and 240 m respectively. Find the other side.
(a) 237 m (b) 181 m (c) 161 m (d) 159 m

17. How many tiles 20 cm by 40 cm will be required to pave the floor of a prayer hall of a room 16 m long and 9 m wide?
(a) 18000 (b) 2700 (c) 1800 (d) 14400

18. If the area of a square be 22050 sq cm. Find the length of diagonal.
(a) 201 cm (b) 220 cm
(c) 211 cm (d) 210 cm

19. If requires 90 g paint for painting a door 12 cm × 9 cm, how much paint is required for painting a similar door 4 cm × 3 cm?
(a) 30 g (b) 27 g (c) 10 g (d) 45 g

20. The area of a rectangular football field is 24200 sq m. It is half as broad as it is long. What is the approx minimum distance a man will cover if he wishes to go from one corner to the opposite one?
(a) 283 m (b) 246 m (c) 576 m (d) 289 m

21. The area of the four walls of a room is 2640 sq m and the length is twice the breadth and the height is given as 11 m. What is the area of the ceiling?
(a) 2800 m^2 (b) 3200 m^2
(c) 320 m^2 (d) none of these

22. If the perimeter of a square and a rectangle are the same, then the areas A and B enclosed by them would satisfy the inequality :
(a) $A > B$ (b) $A \geq B$ (c) $A < B$ (d) $A \leq B$

23. If the perimeter of a rectangle and a square each is equal to 80 cm and the difference of their areas is 100 sq cm, the sides of the rectangle are :
(a) 25 cm, 15 cm (b) 28 cm, 12 cm
(c) 30 cm, 10 cm (d) 35 cm, 15 cm

24. The number of square shaped tin sheets of side 25 cm that can be cut off from a square tin sheet of side 1 m, is :
(a) 4 (b) 40 (c) 16 (d) 400

25. The length and breadth of a rectangular field are 120 m and 80 m respectively. Inside the field, a park of 12 m width is made around the field. The area of the park is :
(a) 2358 m^2 (b) 7344 m^2 (c) 4224 m^2 (d) 3224 m^2

26. A 5m wide lawn is cultivated all along the outside of a rectangular plot measuring 90m × 40 m. The total area of the lawn is :
(a) 1441 m^2 (b) 1400 m^2 (c) 2600 m^2 (d) 420 m^2

27. The length of a rectangle is 2 cm more than its breadth. The perimeter is 48 cm. The area of the rectangle (in cm^2) is :
(a) 96 (b) 128 (c) 143 (d) 144

28. The area of a rectangular field is 52000 m^2. This rectangular area has been drawn on a map to the scale 1 cm to 100 m. The length is shown as 3.25 cm on the map. The breadth of the rectangular field is :
(a) 210 m (b) 150 m
(c) 160 m (d) 123 m

29. If the length of diagonal *BD* of a square *ABCD* is 4.8 cm, the area of the square *ABCD* is :
(a) 9.6 cm^2 (b) 11.52 cm^2
(c) 12.52 cm^2 (d) 5.76 cm^2

30. If the side of square is increased by 20%, then how much per cent does its area get increased?
(a) 40% (b) 20% (c) 44% (d) 24%

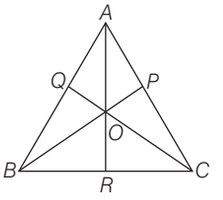
31. The ratio of the area of a square to that of the square drawn on its diagonal is :
(a) 1 : 1 (b) 1 : 2 (c) 1 : 3 (d) 1 : 4

32. The length and breadth of a square are increased by 60% and 40% respectively. The area of the resulting rectangle exceeds the area of the square by :
(a) 224% (b) 24%
(c) 124% (d) 100%

10.4 Triangles

1. Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{height}$ (General formula)
2. Area of a scalene triangle $= \sqrt{s(s-a)(s-b)(s-c)}$
   where s is the semiperimeter of the triangle and a, b and c are the three sides of the triangle, $\left(s = \frac{a+b+c}{2}\right)$
   Scalene triangle : A triangle whose all sides are different in length
3. Area of a right angle triangle $= \frac{1}{2} \times \text{base} \times \text{height}$
   Right angle triangle : A triangle in which two sides are perpendicular
   Also, Hypotenuse $= \sqrt{(\text{base})^2 + (\text{height})^2}$
4. Area of an equilateral triangle
   $= \frac{1}{2} \times \text{side} \times \left(\frac{\sqrt{3}}{2}\text{ side}\right) = \frac{\sqrt{3}}{4} \times (\text{side})^2$
   Equilateral triangle : A triangle in which all the three sides are equal also all the three internal angles are equal
   Height of an equilateral triangle $(h) = \frac{\sqrt{3}}{2} \times \text{side}$

An important property : In an equilateral Δ perpendiculars drawn from all the three vertices intersect each other in the ratio of 2 : 1 from the vertex to the base.

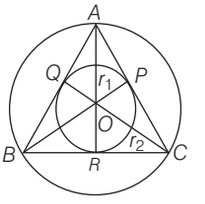


$\therefore \quad \frac{AO}{OR} = \frac{BO}{OP} = \frac{CO}{OQ} = \frac{2}{1} = \frac{r_2}{r_1}$

$OP = OQ = OR$, all are the inradii

$OA = OB = OC$,

all are the circumradii



NOTE Radii means more than one radius. *O* is the centre of two circles. Circle *PQR* is called as incircle (touching the sides) and circle *ABC* is called as circumcircle (touching the vertices)

$\therefore \quad$ Inradius $= \frac{\text{side}}{2\sqrt{3}}$ and Circumradius $= \frac{\text{side}}{\sqrt{3}}$

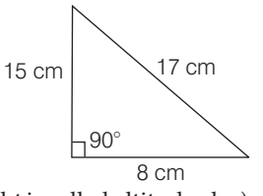
Exp. 1) The base of a right angled triangle is 8 cm and hypotenuse is 17 cm. Find its area.

Solution Hypotenuse $= \sqrt{(\text{base})^2 + (\text{altitude})^2}$

$(17) = \sqrt{64 + (x)^2}$

$\Rightarrow \quad 289 = 64 + x^2$

$\Rightarrow \quad x^2 = 225 \Rightarrow x = 15$ cm



$\therefore \quad$ Area $= \frac{1}{2} \times \text{base} \times \text{altitude}$

(height is called altitude also)

$= \frac{1}{2} \times 8 \times 15 = 60\ cm^2$

Exp. 2) Find the area of an equilateral triangle whose side is $4\sqrt{3}$ cm.

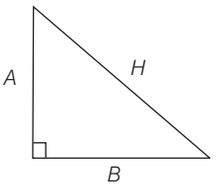
Solution Area of an equilateral triangle $= \frac{\sqrt{3}}{4} \times (\text{side})^2$

$$= \frac{\sqrt{3}}{4} \times (4\sqrt{3})^2 = \frac{\sqrt{3}}{4} \times 48 = 12\sqrt{3} \text{ cm}^2$$

Exp. 3) The area of a right angled triangle is 24 cm^2 and the length of its hypotenuse is 10 cm. The length of the shorter leg is :

(a) 5 cm (b) 4 cm
(c) 6 cm (d) 3 cm

Solution $H^2 = A^2 + B^2 = 100 \text{ cm}^2$



$$\text{Area} = \frac{1}{2} \times A \times B$$

$$\Rightarrow \quad \frac{1}{2} A \times B = 24 \text{ cm}^2$$

$$\Rightarrow \quad A \times B = 48 \text{ cm}^2$$

$$\therefore \quad (A + B)^2 = A^2 + B^2 + 2AB$$

$$(A + B)^2 = 196 \Rightarrow A + B = 14 \qquad \ldots(i)$$

Again $(A - B)^2 = A^2 + B^2 - 2AB$

$$\Rightarrow \quad (A - B)^2 = 4 \Rightarrow A - B = 2 \qquad \ldots(ii)$$

Therefore by solving eqs. (i) and (ii) ,we get

$$A = 8 \quad \text{and} \quad B = 6$$

Therefore, the shorter leg is 6 cm.

Alternatively Go through options.

$$\because \quad \frac{1}{2} \times A \times B = 24 \Rightarrow A \times B = 48$$

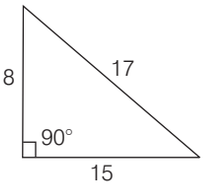
Let us assume $B = 6$, then $A = \frac{48}{6} = 8$

Now, $A^2 + B^2 = 100, (8)^2 + (6)^2 = 100 \Rightarrow 100 = 100$

Hence, choice (c) is correct.

Exp. 4) Find the area and perimeter of a triangle whose sides are 17 cm, 8 cm and 15 cm long.

Solution Since, $8^2 + 15^2 = 17^2$



$$\Rightarrow \quad 64 + 225 = 289$$

$$289 = 289$$

Hence, it is a right angled triangle.

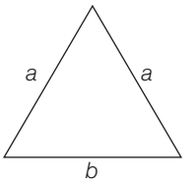
$$\therefore \quad \text{Area} = \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \times 15 \times 8 = 60 \text{ cm}^2$$

and perimeter $= 8 + 15 + 17 = 40$ cm

Exp. 5) Find the area and perimeter of an isosceles triangle whose equal sides are 5 cm each and base is 6 cm.

Solution Area of an isosceles triangle

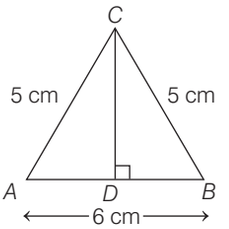


$$= \frac{b}{4} \sqrt{4a^2 - b^2}$$

$$= \frac{6}{4} \sqrt{4 \times 25 - 36}$$

$$= \frac{3}{2} \times \sqrt{64} = 12 \text{ cm}^2$$

Alternatively We know that the altitude CD bisects the base AB in the isosceles triangle ABC.



$$\therefore \quad AD = BD = 3 \text{ cm}$$

Using Pythagoras theorem in ΔADC we have,

$$CD^2 = AC^2 - AD^2$$

$$\therefore \quad CD = 4 \text{ cm}$$

$$\therefore \text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$$

Also, the perimeter of triangle $= 5 + 5 + 6 = 16$ cm

Introductory Exercise 10.2

1. What is the area of the triangle whose sides are 84 m, 80 m and 52 m?
(a) 1620 sq. m (b) 2016 sq. m
(c) 1818 sq. m (d) none of these

2. Two poles 15 m and 30 m high stand upright in a play ground. If their feet be 36 m apart, find the distance between their tops.
(a) 41 m
(b) 36 m
(c) 39 m
(d) none of the above

3. The sides of a triangle are 25 m, 39 m and 56 m respectively. Find the perpendicular distance from the vertex opposite to the side 56 m.
(a) 15 m (b) 16.5 m
(c) 18.6 m (d) 21 m

4. ABC is a triangle and D, E, F are the mid-points of the sides BC, CA, AB respectively. The ratio of the areas of ΔABC and ΔDEF is :
(a) 4 : 1 (b) 5 : 1
(c) 3 : 1 (d) can't be determined

5. The integral base of an isosceles triangle can be whose area is 60 cm^2 and the length of one of the equal sides is 13 cm :
(a) 20 cm (b) 10 cm
(c) 16 cm (d) data insufficient

6. A ladder is resting with one end in contact with the top of a wall of height 60 m and the other end on the ground is at a distance of 11 m from the wall. The length of the ladder is :
(a) 61 m (b) 71 m
(c) 87 m (d) none of these

7. The base of a triangular field is three times its height. If the cost of cultivating the field is ₹ 36.72 per hectare is ₹ 495.72, find the height and base of the triangular field.
(a) 480 m, 1120 m (b) 400 m, 1200 m
(c) 300 m, 900 m (d) 250 m, 650 m

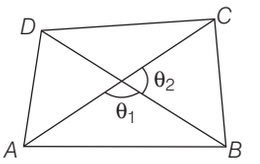
8. If every side of a triangle is doubled, then increase in area of the triangle is :
(a) 200% (b) 300%
(c) 400% (d) none of these

9. If the altitude of an equilateral triangle is $2\sqrt{3}$, then its area is :
(a) $4\sqrt{3}$ cm^2 (b) $12\sqrt{3}$ cm^2
(c) $\frac{8}{\sqrt{3}}$ cm^2 (d) none of these

10. If the perimeter of an equilateral triangle and a square is same and the area of equilateral triangle is *P* and the area of square is *Q*, then :
(a) $P < Q$ (b) $P \le Q$
(c) $P > Q$ (d) $P \ge Q$

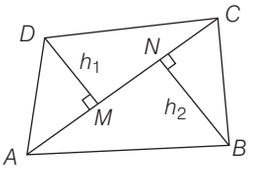
10.5 **Parallelogram, Rhombus and Trapezium**

1. (a) Area of parallelogram = base (b) × height (h)
   (b) Area of parallelogram
   = product of any two adjacent sides
   × sine of the included angle
2. Perimeter = 2 (sum of any two adjacent sides)
3. (a) Area of rhombus $= \frac{1}{2} \times$ (product of diagonals)
   $= \frac{1}{2} \times d_1 d_2$
   (b) Area of rhombus = product of adjacent sides
   × sine of the angle included by them
4. Perimeter of rhombus = 4 × side
5. Area of a trapezium $= \frac{1}{2} \times$ sum of parallel sides × height
   height → distance between the two parallel sides
6. Perimeter of trapezium = sum of all the four sides
7. (a) Area of quadrilateral $= \frac{1}{2} \times$ product of diagonal
   × sine of the angle between them
   ($AC = d_1$ and $BD = d_2$)



$\therefore$ Area $= \frac{1}{2} d_1 d_2 \sin\theta_1 = \frac{1}{2} d_1 d_2 \sin\theta_2$

(b) Area $= \frac{1}{2} \times$ diagonal × sum of the perpendiculars drawn from the opposite vertices on it.



$= \frac{1}{2} d \times (h_1 + h_2)$

where, $MD = h_1$ and $BN = h_2$ and $AC = d$

Introductory Exercise 10.3

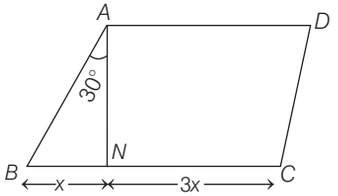
1. The adjacent sides of a parallelogram are 6 cm and 8 cm and the angle between them is 30°. What is the area of the parallelogram?
(a) 24 cm^2 (b) 12 cm^2
(c) 40 cm^2 (d) $24\sqrt{3}$ cm^2

2. A parallelogram has sides 30 cm and 20 cm and one of its diagonal is 40 cm long. Then its area is :
(a) $75\sqrt{5}$ cm^2 (b) 245 cm^2
(c) $150\sqrt{15}$ cm^2 (d) 300 cm^2

3. The distance of a 24 cm long side of a parallelogram from the opposite side is 22 cm. The area of the parallelogram is :
(a) 264 cm^2 (b) 246 cm^2
(c) 460 cm^2 (d) 528 cm^2

4. The two adjacent sides of a parallelogram are 25 cm and 40 cm respectively. The altitude drawn on the longer side is 18 cm, then the area of the parallelogram is :
(a) 450 cm^2 (b) 720 cm^2
(c) 500 cm^2 (d) none of these

5. In the adjoining figure, the ratio of the areas of the parallelogram *ABCD* and that of triangle *ABN* is :



(a) 6 : 1 (b) 5 : 1
(c) 4 : 1 (d) 8 : 1

6. If the perimeter of a rhombus is $4p$ and lengths of its diagonals are a and b, then its area is :
(a) $\frac{a}{b}$ (b) $\frac{ab}{2}$
(c) ab/p (d) $p\,(a^2 + b^2)$

7. The ratio of the lengths of the diagonal of a rhombus is 2 : 5. Then, the ratio of the area of the rhombus to the square of the shorter diagonal :
(a) 5 : 4 (b) 5 : 2
(c) 2 : 5 (d) none of these

8. Area of a rhombus is 256 cm^2. One of the diagonal is half of the other diagonal. The sum of the diagonals is :
(a) 38 cm (b) 48 cm
(c) 28 cm (d) 56 cm

9. The lengths of two parallel sides of a trapezium are 30 cm and 50 cm and its height is 16 cm. Its area is :
(a) 460 cm^2 (b) 750 cm^2
(c) 320 cm^2 (d) 640 cm^2

10. *ABCD* is a trapezium in which $AB \parallel CD$ and $AB = 2CD$. If its diagonals intersect each other at *O*, then ratio of areas of triangles *AOB* and *COD* is :
(a) 1 : 4 (b) 1 : 2
(c) 4 : 1 (d) 2 : 1

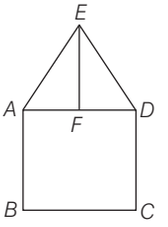
11. The area of a trapezium is 441 cm^2 and the ratio of parallel sides is 5 : 9. Also the perpendicular distance between them is 21 cm, the longer of parallel sides is :
(a) 36 cm (b) 27 cm
(c) 18 cm (d) 28 cm

12. The cross-section of a canal is in the shape of a trapezium and the area of cross-section is 360 m^2. If the canal is 12 m wide at the top and 8 m wide at the bottom the depth of the canal is :
(a) 36 m (b) 180 m
(c) 45 m (d) none of these

13. The area of a hexagon whose one side is 4 m, is :
(a) $6\sqrt{3}$ m^2 (b) $24\sqrt{3}$ m^2
(c) $42\sqrt{3}$ m^2 (d) 24 m^2

14. *ABCD* is a quadrilateral $AC = 19$ cm. The lengths of perpendiculars from *B* and *D* on *AC* are 5 cm and 7 cm respectively. Then, the area of *ABCD* (in cm^2) is :
(a) 162 (b) 144
(c) 228 (d) 114

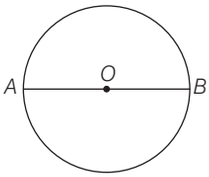
15. *ABCD* is a square, $AC = BD = 4\sqrt{2}$ cm, $AE = DE = 2.5$ cm. Find the area of the adjoining figure *ABCDE*.
(a) 19 cm^2
(b) 22 cm^2
(c) 17 cm^2
(d) none of the above



10.6 Circles

1. Area of a circle $= \pi R^2$ ($R \rightarrow$ radius of the circle)
2. Circumference of the circle $= 2\pi R$
3. Length of an arc $= 2\pi R\left(\frac{\theta}{360°}\right)$
4. Area of a sector $= \pi R^2\left(\frac{\theta}{360°}\right) = \frac{1}{2}(\text{arc} \times R)$
5. Area of segment $= \pi R^2\left(\frac{\theta}{360°}\right) - \frac{R^2}{2}\sin\theta$

Diameter $= 2 \times$ radius



$OB = OA \rightarrow$ radius

$AB \rightarrow$ diameter

Exp. 1) The radius of a circular wheel is $1\frac{3}{4}$ m. How many revolutions will it make in travelling 11 km?

Solution Total distance (travelled) $= 11$ km $= 11000$ m

Distance travelled in one revolution

= circumference of the wheel

$= 2 \times \pi \times r = 2 \times \frac{22}{7} \times \frac{7}{4} = 11$ m

$\therefore$ Number of revolutions in 11 km $= \frac{11000}{11}$

$= 1000$ revolution

Exp. 2) It takes 13.5 mL to paint the surface of the circular sheet of radius 17 cm. How much paint is required to paint a similar circular sheet with double the radius?

Solution Ratio of radii $= 1 : 2$

$\therefore$ Ratio of areas $= 1 : 4$

$\left(\text{Since } \frac{\text{Area of } C_1}{\text{Area of } C_2} = \frac{\pi r_1^2}{\pi r_2^2} = \frac{1 \times 1}{2 \times 2} = \frac{1}{4}\right)$

$\therefore$ Quantity of required paint is 4 times.

Thus we need $4 \times 13.5 = 54$ mL paint.

Exp. 3) A circular road runs round a circular garden. If the difference between the circumference of the outer circle and the inner circle is 44 m. Find the width of the road.

Solution Let R and r be radii of outer circle and inner circle respectively.

$\therefore$ Width of the road $= R - r$

$\therefore \quad 2\pi R - 2\pi r = 44\text{m} \Rightarrow 2\pi(R - r) = 44$ m

$\Rightarrow \quad (R - r) = 7$ m $\quad \left(\because \pi = \frac{22}{7}\right)$

Exp. 4) The radius of a circle is 5 m. What is the radius of another circle whose area is 25 times that of the first?

Solution Ratio of areas = (ratio of radii)^2

$\frac{25}{1} = (\text{ratio of radii})^2 \Rightarrow \text{Ratio of radii} = \frac{5}{1}$

Therefore radius of another circle is 5 times.

Hence, the required radius = 25 m

Exp. 5) What is the radius of a circle whose area is equal to the sum of the areas of two circles whose radii are 20 cm and 21 cm?

Solution

$\pi R^2 = \pi r_1^2 + \pi r_2^2$

$\pi R^2 = \pi(r_1^2 + r_2^2)$

$R^2 = (400 + 441)$

$R^2 = 841 \Rightarrow R = 29$ cm

Exp. 6) In a circle of radius 28 cm, an arc subtends an angle of 108° at the centre.

(a) Find the area of the sector.

(b) Find the length of the arc.

Solution (a) Area of the sector $= \pi r^2\left(\frac{\theta}{360°}\right)$

$= \frac{22}{7} \times 28 \times 28 \times \frac{108°}{360°}$

$= 22 \times 4 \times 28 \times \frac{3}{10} = 739.2$ cm

(b) Length of the arc $= 2\pi r\left(\frac{\theta}{360°}\right)$

$= 2 \times \frac{22}{7} \times 28 \times \frac{108}{360°} = 52.8$ cm

Introductory Exercise 10.4

1. If the circumference of a circle is 704 cm, then its area is :
(a) 49324 m^2 (b) 39424 m^2
(c) 3672 cm^2 (d) 39424 cm^2

2. If the circumference of a circle is 4.4 m, then the area of the circle (in m^2) is
(a) $49/\pi$ (b) 49π
(c) 4.9π (d) none of these

3. A circular wire of radius 4.2 m is cut and bent in the form of a rectangle whose longer side is 20% more than its shorter side. The longer side of the rectangle is :
(a) 7.2 m (b) 72 cm
(c) 8 m (d) none of these

4. The inner circumference of a circular path around a circular lawn is 440 m. What is the radius of the outer circumference of the path, if the path is 14 m wide?
(a) 96 m (b) 84 m
(c) 70 m (d) 88 m

5. The sum of the radius and the circumference of a circle is 51 cm. The area of the circle is :
(a) 151 cm (b) 152 cm
(c) 154 cm (d) data insufficient

6. The difference between the circumference and the diameter of the circle is 15 m. What is the area of the circle?
(a) 225 m^2 (b) 165 m^2
(c) 156 m^2 (d) none of these

7. The radius of a circle is increased by 2 cm from 5 cm to 7 cm. What is the percentage change in area of the circle?
(a) 96% (b) 35%
(c) 70% (d) 74%

8. If the circumference of a circle is increased by 20%, then its area will be increased by :
(a) 44% (b) 32%
(c) 40% (d) none of these

9. The area of a circular field is 124.74 hectares. The cost of fencing it at the rate of 80 paise per metre is
(a) ₹ 3168 (b) ₹ 1584
(c) ₹ 1729 (d) none of these

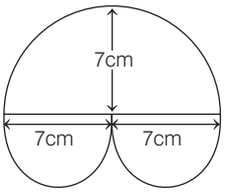
10. Eldeco Housing Pvt. Ltd purchased a circular plot of land for ₹ 158400 at the rate of 1400 per sq. metre. The radius of the plot is :
(a) 5 m (b) 6 m
(c) 7 m (d) 14 m

11. A figure consists of a square of side 'a' m with semicircles drawn on the outside of the square. The area (in m^2) of the figure so formed will be :
(a) $a^2(\pi + 1)$ (b) $a^2\left(\pi + \frac{1}{4}\right)$
(c) $a^2 + \frac{\pi a^2}{2}$ (d) none of these

12. The length of a rope by which a buffalo must be tethered so that she may be able to graze a grassy area of 2464 sq m is :
(a) 35 m (b) 27 m
(c) 24 m (d) 28 m

13. A circle of radius 'a' is divided into 6 equal sectors. An equilateral triangle is drawn on the chord of each sector to lie outside the circle. Area of the resulting figure is :
(a) $3a^2(\pi + \sqrt{3})$ (b) $3\sqrt{3}\ a^2$
(c) $3(a^2\sqrt{3} + \pi)$ (d) $\frac{3\sqrt{3}\pi a^2}{2}$

14. In the following figure, the area in (cm^2) is :

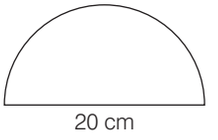


(a) 115.5 (b) 228.5
(c) 154 (d) none of these

15. If a piece of wire 25 cm long is bent into an arc of a circle subtending an angle of 75° at the centre, then the radius of the circle (in cm) is :
(a) $\frac{\pi}{120}$ (b) $\frac{60}{\pi}$
(c) 60π (d) none of these

16. Four horses are tethered at four corners of a square plot of 42 m so that they just cannot reach one another. The area left ungrazed is :
(a) 378 m^2 (b) 438 m^2
(c) 786 m^2 (d) none of these

17. The circumference of the following figure is :



(a) $(20 + 10\pi)$ (b) 20π
(c) 10π (d) 30π

18. The area of a minor sector subtending the central angle at the centre 40° is 8.25 cm^2. What is the area of the remaining part (i.e. major sector) of the circle?
(a) 82.5 cm^2 (b) 74.25 cm^2
(c) 66 cm^2 (d) none of these

19. The area of a sector of a circle of radius 8 cm, formed by an arc of length 5.6 cm, is :
(a) 22.4 cm^2 (b) 2.24 cm^2
(c) 56 cm^2 (d) none of these

20. How long will a man take to go, walking at 13.2 km/h, round a circular garden of 700 m radius?
(a) 12 min (b) 30 min
(c) 20 min (d) none of these

21. What is the radius of circular field whose area is equal to the sum of the areas of three smaller circular fields of radii 8 m, 9 m and 12 m respectively?
(a) 17 m (b) 20 m
(c) 21 m (d) 29 m

22. A rope by which a calf is tied is decreased from 23 m to 12 m. What is the decrease in area to be grazed by it?
(a) 1110 m^2 (b) 1210 m^2
(c) 1120 m^2 (d) 1221 m^2

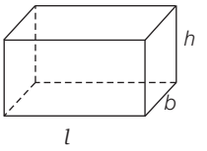
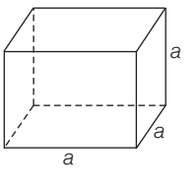
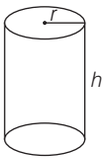
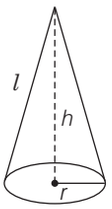
23. A wire is bent in the form of a square of side 66 m. It is cut and again bent in the form of a circle. The diameter of this circle is :
(a) 42 m (b) 84 m
(c) 21 m (d) none of these

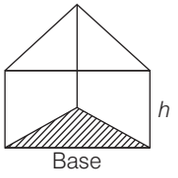
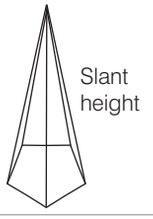
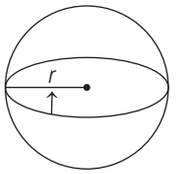
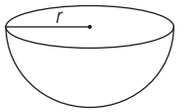
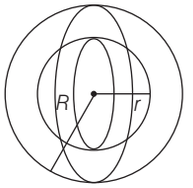
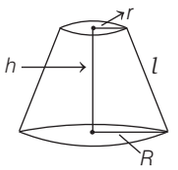
24. A wire is in the form of a circle of radius 42 m is cut and again bent in the form of a square. What is the diagonal of the square?
(a) 66 m (b) $66\sqrt{3}$ m
(c) $66\sqrt{2}$ m (d) none of these

25. If the driving wheel of a bicycle makes 560 revolutions in travelling 1.1 km. Find the diameter of the wheel.
(a) 31.5 cm (b) 30.5 cm
(c) 62.5 cm (d) none of these

10.7 **3-D Figures**

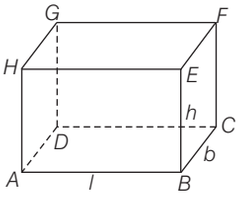
3-D Figures are shown below

Name	Figure	Nomenclature	Volume	Curved/Lateral surface area	Total surface area
Cuboid		$l \rightarrow$ length $b \rightarrow$ breadth $h \rightarrow$ height	lbh	$2(l+b)h$	$2(lb+bh+hl)$
Cube		$a \rightarrow$ edge/side	a^3	$4a^2$	$6a^2$
Right circular cylinder		$r \rightarrow$ radius of base $h \rightarrow$ height of the cylinder	$\pi r^2 h$	$2\pi rh$	$2\pi r(h+r)$
Right circular cone		$r \rightarrow$ radius $h \rightarrow$ height $l \rightarrow$ slant height $l = \sqrt{r^2 + h^2}$	$\frac{1}{3}\pi r^2 h$	πrl	$\pi r(l+r)$

Name	Figure	Nomenclature	Volume	Curved/Lateral surface area	Total surface area
Right triangular prism		—	area of base × height	perimeter of base × height	lateral surface area + 2 (area of base)
Right pyramid		—	$\frac{1}{3}$ × area of the base × height	$\frac{1}{2}$ × perimeter of the base × slant height	lateral surface area + area of the base
Sphere		$r \rightarrow$ radius	$\frac{4}{3}\pi r^3$	—	$4\pi r^2$
Hemisphere		$r \rightarrow$ radius	$\frac{2}{3}\pi r^3$	$2\pi r^2$	$3\pi r^2$
Spherical shell		$r \rightarrow$ inner radius $R \rightarrow$ outer radius	$\frac{4}{3}\pi [R^3 - r^3]$	—	$4\pi [R^2 + r^2]$
Frustum of a cone		—	$\frac{\pi}{3} h (r^2 + Rr + R^2)$	$\pi (r + R) l$	lateral surface area + $\pi [R^2 + r^2]$

10.8 Cuboid and Cube

Cuboid : A cuboid has 6 faces, 12 edges, 8 vertices and 4 diagonals.



Faces :
$ABCD$, $EFGH$, $ABEH$, $CFGD$, $BCFE$, $ADGH$

Edges : $AB, BC, CD, AD, EF, FG, GH, EH, BE, CF, DG, AH$

Vertices : A, B, C, D, E, F, G, H

Diagonals : AF, BG, CH, DE

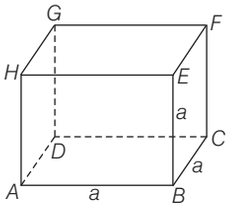
Formulae : Volume $= l \times b \times h$

($l \rightarrow$ length, $b \rightarrow$ breadth, $h \rightarrow$ height)

$$\text{Total surface area} = 2(lb + bh + hl)$$

$$\text{Diagonal } (d) = \sqrt{l^2 + b^2 + h^2}$$

Cube : A cube has 6 equal faces, 12 equal edges, 8 vertices and 4 equal diagonals.



Formulae :

$$\text{Volume} = (a)^3$$

$$\text{Total surface area} = 6(a)^2$$

($a \rightarrow$ edge of the cube)

$$\text{Diagonal } (d) = a\sqrt{3}$$

Euler's Theorem $\rightarrow (V + F) = (E + 2)$; where $V \rightarrow$ number of vertices, $F \rightarrow$ number of faces, $E \rightarrow$ number of edges

Exp. 1) The dimensions of a cuboid are 16 cm, 18 cm and 24 cm. Find :

(a) volume (b) surface area (c) diagonal

Solution (a) Volume $= l \times b \times h = 16 \times 18 \times 24 = 6912 \text{ cm}^3$ (b) Surface area $= 2(lb + bh + hl)$

$= 2(16 \times 18 + 18 \times 24 + 24 \times 16)$

$= 2208 \text{ cm}^2$

(c) Diagonal $= \sqrt{l^2 + b^2 + h^2}$

$= \sqrt{16^2 + 18^2 + 24^2} = \sqrt{1156} = 34 \text{ cm}$

Exp. 2) Edge of a cube is 5 cm. Find :

(a) volume (b) surface area (c) diagonal

Solution Volume $= a^3 = (5)^3 = 125 \text{ cm}^3$

Surface area $= 6a^2 = 6 \times (5)^2 = 150 \text{ cm}^2$

Diagonal $= a\sqrt{3} = 5\sqrt{3} = 8.660 = 8.66 \text{ cm}$

Exp. 3) Three cubes of volumes, 1 cm^3, 216 cm^3 and 512 cm^3 are melted to form a new cube. What is the diagonal of the new cube?

Solution Volume of new cube $= 1 + 216 + 512 = 729 \text{ cm}^3$

$\therefore$ Edge of new cube $= \sqrt[3]{729} = 9 \text{ cm}$

$\therefore$ Surface area $= 6a^2 = 6 \times (9)^2 = 486 \text{ cm}^2$

$\therefore$ Diagonal of the new cube $= a\sqrt{3} = 9\sqrt{3}$

$= 15.6 \text{ cm}$ (approx.)

Exp. 4) The surface area of a cube is 864 cm^2. Find its volume.

Solution $6a^2 = 864 \Rightarrow a^2 = 144 \quad \Rightarrow a = 12 \text{ cm}$

$\therefore \quad a^3 = (12)^3 = 1728 \text{ cm}^3$

Exp. 5) Find the length of the longest pole that can be placed in a room 30 m long, 24 m broad and 18 m high.

Solution $d = \sqrt{l^2 + b^2 + h^2}$

$d = \sqrt{900 + 576 + 324} \Rightarrow d = 30\sqrt{2} \text{ m}$

Exp. 6) A brick measures 20 cm $\times$ 10 cm $\times$ 7.5 cm. How many bricks will be required for a wall 20 m $\times$ 2 m $\times$ 0.75 m?

Solution $\text{Number of bricks} = \dfrac{\text{total volume of a wall}}{\text{volume of one brick}}$

$= \dfrac{20 \times 2 \times 0.75 \times 100 \times 100 \times 100}{20 \times 10 \times 7.5} = 20000$

Introductory Exercise 10.5

1. A cube of metal, each edge of which measures 4 cm, weighs 400 kg. What is the length of each edge of a cube of the same metal which weighs 3200 kg?
(a) 64 cm (b) 8 cm
(c) 2 cm (d) none of these

2. The length of a tank is thrice that of breadth, which is 256 cm deep and holds 3000 L water. What is the base area of the tank? (1000 L = 1 cubic metre)
(a) 111775 m^2 (b) 1171.875 m^2
(c) 1.171875 m^2 (d) none of these

3. The three co-terminus edges of a rectangular solid are 36 cm, 75 cm and 80 cm respectively. Find the edge of a cube which will be of the same capacity.
(a) 60 cm (b) 52 cm
(c) 46 cm (d) none of these

4. A tank 10 m long and 4 m wide is filled with water. How many litres of water must be drawn off to make the surface sink by 1 m? (1000 L = 1 cubic metre)
(a) 20 kilolitre (b) 40 kilolitre
(c) 50 kilolitre (d) none of these

5. How many cubes each of surface area 24 sq. dm can be made out of a metre cube, without any wastage?
(a) 75 (b) 250
(c) 125 (d) 62

6. Three cubes of metal, whose edges are 3 cm, 4 cm and 5 cm respectively are melted to form a new cube. What is the surface area of the new cube?
(a) 216 cm^2 (b) 56 cm^2
(c) 36 cm^2 (d) none of these

7. A lid of rectangular box of sides 39.5 cm by 9.35 cm is sealed all around with tape such that there is an overlapping of 3.75 cm of the tape. What is the length of the tape used?
(a) 111.54 cm (b) 101.45 cm
(c) 110.45 cm (d) none of these

8. A cistern from inside is 12.5 m long, 8.5 m broad and 4 m high and is open at top. Find the cost of cementing the inside of a cistern at ₹ 24 per sq m.
(a) ₹ 6582 (b) ₹ 8256
(c) ₹ 7752 (d) ₹ 8752

9. 250 men took a dip in a water tank at a time, which is 80 m × 50 m. What is the rise in the water level if the average displacement of 1 man is 4 m^3?
(a) 22 cm (b) 25 cm (c) 18 cm (d) 30 cm

10. The edge of a cube is increased by 100%, the surface area of the cube is increased by :
(a) 100% (b) 200% (c) 300% (d) 400%

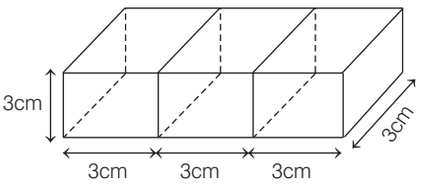
11. The edge of a cube is doubled. What will be the new volume?
(a) 2 times (b) 3 times
(c) 4 times (d) 8 times

12. The external dimensions of a wooden box closed at both ends are 24 cm, 16 cm and 10 cm respectively and thickness of the wood is 5 mm. If the empty box weighs 7.35 kg, find the weight of 1 cubic cm of wood.
(a) 10 g (b) 12.5 g
(c) 27 g (d) 15 g

13. The internal dimensions of a tank are 12 dm, 8 dm and 5 dm. How many cubes each of edge 7 cm can be placed in the tank with faces parallel to the sides of the tank. Find also, how much space is left unoccupied.
(a) 35; 113 dm^3 (b) 1313; 31.13 dm^3
(c) 1309; 31.013 dm^3 (d) 1309; 13.31 dm^3

14. The length, breadth and height of box are 2 m, 1.5 m and 80 cm respectively. What would be the cost of canvas to cover it up fully, if one square metre of canvas costs ₹ 25.00?
(a) ₹ 260 (b) ₹ 290
(c) ₹ 285 (d) none of these

15. Three cubes each of edge 3 cm long are placed together as shown in the adjoining figure. Find the surface area of the cuboid so formed :



(a) 182 sq cm (b) 162 sq cm
(c) 126 sq cm (d) none of these

16. A room is 36 m long, 12 m wide and 10 m high. It has 6 windows, each 3 m × 2.5 m; one door 9.5 m × 6 m and one fire chimney 4 m × 4.5 m. Find the expenditure of papering its walls at the rate of 70 paise per metre, if the width of the paper is 1.2 m.
(a) ₹ 490 (b) ₹ 690
(c) ₹ 1000 (d) none of these

17. A school hall has the dimensions 30 m, 12 m by 6 m. Find the number of children who can be accommodated, if each child should get 8 m^3 of space.
(a) 240 (b) 270
(c) 250 (d) 150

18. When each side of a cube is increased by 2 cm, the volume is increased by 1016 cm^3. Find the side of the cube. If each side of it is decreased by 2 cm, by how much will the volume decrease?
(a) 12 cm, 729 cm^3
(b) 8 cm, 512 cm^3
(c) 9 cm, 729 cm^3
(d) 12 cm, 728 cm^3

19. Three equal cubes are placed adjacently in a row. Find the ratio of the total surface area of the resulting cuboid to that of the sum of the total surface areas of the three cubes.
(a) 5 : 7 (b) 7 : 9
(c) 9 : 7 (d) none of these

20. A hollow square shaped tube open at both ends is made of iron. The internal square is of 5 cm side and the length of the tube is 8 cm. There are 192 cm^3 of iron in the tube. Find its thickness.
(a) 2 cm (b) 0.5 cm
(c) 1 cm (d) can't be determined

21. A cube of 11 cm edge is immersed completely in a rectangular vessel containing water. If the dimensions of base are 15 cm and 12 cm. Find the rise in water level in the vessel.
(a) 6.85 cm (b) 7 cm
(c) 7.31 cm (d) 7.39 cm

22. A rectangular tank 25 cm long and 20 cm wide contains water to a depth of 5 cm. A metal cube of side 10 cm is placed in the tank so that one face of the cube rests on the bottom of the tank. Find how many litres of water must be poured into the tank so as to just cover the cube.
(a) 1 L (b) 1.5 L
(c) 2 L (d) 2.5 L

23. A rectangular block has length 10 cm, breadth 8 cm and height 2 cm. From this block, a cubical hole of side 2 cm is drilled out. Find the volume and the surface area of the remaining solid.
(a) 152 cm^3, 512 cm^2 (b) 125 cm^3, 215 cm^2
(c) 152 cm^3, 240 cm^2 (d) 125 cm^3, 512 cm^2

24. A rectangular tank of dimensions 24 m × 12 m × 8 m is dug inside a rectangular field 600 m long and 200 m broad. The earth taken out is evenly spread over the field. By how much will the level of the field rise?
(a) 1.925 cm (b) 0.02 m
(c) 0.2 cm (d) none of these

25. How many bricks (number near to next hundred) will be required to build a wall 30 m long, 30 cm thick and 5 m high with a provision of 2 doors, each 4 m × 2.5 m and each brick being 20 cm × 16 cm × 8 cm when one-ninth of the wall is filled with lime?
(a) 13500 bricks (b) 13600 bricks
(c) 20050 bricks (d) 18500 bricks

26. A rectangular water reservoir is 15 m by 12 m at the base. Water flows into it through a pipe whose cross-section is 5 cm by 3 cm at the rate of 16 m per second. Find the height to which the water will rise in the reservoir in 25 min.
(a) 0.2 m (b) 2 cm
(c) 0.5 m (d) none of these

27. Which of the following pairs is not correctly matched?

Geometrical objects	Number of vertices
(A) Tetrahedron	4
(B) Pyramid with rectangular base	5
(C) Cube	6
(D) Triangle	3

(a) (A) (b) (B)
(c) (C) (d) (D)

28. If the length of diagonal of a cube is $6\sqrt{3}$ cm, then the length of its edge is :
(a) 2 cm (b) 3 cm
(c) 6 cm (d) $\frac{36}{\sqrt{3}}$ cm

29. The length of longest pole that can be placed on the floor of a room is 12 m and the length of longest pole that can be placed in the room is 15 m. The height of the room is :
(a) 3 m (b) 6 m
(c) 9 m (d) none of these

30. The sum of length, breadth and depth of a cuboid is 12 cm and its diagonal is $5\sqrt{2}$ cm. Its surface area is
(a) 152 cm^2 (b) 94 cm^2
(c) 108 cm^2 (d) $60\sqrt{2}$ cm^2

31. The volume of a wall, 3 times as high as it is broad and 8 times as long as it is high, is 36.864 m^3. The height of the wall is :
(a) 1.8 m (b) 2.4 m
(c) 4.2 m (d) none of these

32. If the areas of 3 adjacent sides of a cuboid are x, y, z respectively, then the volume of the cuboid is :
(a) xyz (b) $\sqrt{xyz}$
(c) $3xyz$ (d) none of these

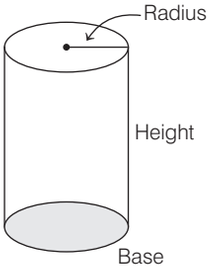
10.9 Cylinder and Cone

Cylinder :

Volume = base area × height = $\pi r^2 h$

Curved surface area

= perimeter × height = $2\pi rh$

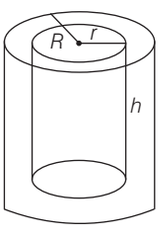


Total surface of the cylinder

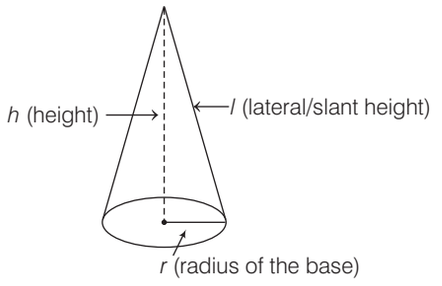
= curved surface area + 2 (base area)

$= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$

Volume of a hollow cylinder = $\pi h\,(R^2 - r^2)$



Cone :



$$\text{Volume } (V) = \frac{1}{3} \times \text{base area} \times \text{height}$$

$$= \frac{1}{3} \pi r^2 h$$

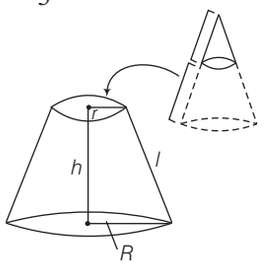
Curved surface area $= \pi rl$

Total surface area $= \pi rl + \pi r^2 = \pi r\,(l + r)$

Frustum of a cone :

Volume of the frustum of a cone

$$= \frac{\pi}{3} h\,(r^2 + Rr + R^2)$$



Curved/Lateral surface area of the frustum of cone

$$= \pi l\,(r + R)$$

Exp. 1) The base radius of a cylinder is 14 cm and its height is 30 cm. Find :

(a) volume (b) curved surface area

(c) total surface area

Solution (a) Volume of cylinder

$$= \pi r^2 \times h = \frac{22}{7} \times 14 \times 14 \times 30$$

$$= 18480 \text{ cm}^3$$

(b) Curved surface area $= 2\pi rh = 2 \times \frac{22}{7} \times 14 \times 30$

$$= 2640 \text{ cm}^2$$

(c) Total surface area $= 2\pi r\,(h + r)$

$$= 2 \times \frac{22}{7} \times 14\,(30 + 14)$$

$$= 2 \times \frac{22}{7} \times 14 \times 44 = 3872 \text{ cm}^2$$

Exp. 2) How many cubic metres of earth must be dug to make a well 14 m deep and 4 m in diameter?

Solution Earth to be dugout from the well

= volume of the cylindrical well

$$= \pi r^2 h = \frac{22}{7} \times 2 \times 2 \times 14$$

$$= 176 \text{ m}^3$$

Exp. 3) A hollow cylindrical tube open at both ends is made of iron 2 cm thick. If the external diameter be 50 cm and the length of the tube is 210 cm, find the number of cubic cm of iron in it.

Solution External radius $(R) = 25$ cm

Internal radius $(r) = (25 - 2) = 23$ cm

Volume of iron $= \pi h\,(R^2 - r^2)$

$$= \frac{22}{7} \times 210 \times (25^2 - 23^2)$$

$$= 63360 \text{ cm}^3$$

Exp. 4) A well with 14 m inside diameter is dugout 15 m deep. The earth taken out of it has been evenly spread all around it to a width of 21 m to form an embarkment. What is the height of the embarkment?

Solution Area of embarkment $\times$ height of embarkment

= volume of earth dugout

$$\pi\,(R^2 - r^2) \times h = \pi \times 7 \times 7 \times 15$$

$$\Rightarrow \quad (28^2 - 7^2)\,h = 7 \times 7 \times 15$$

$$\Rightarrow \quad (35 \times 21) \times h = 7 \times 7 \times 15$$

$$\Rightarrow \quad h = 1 \text{ m}$$

Exp. 5) A cylindrical cistern whose diameter is 21 cm is partly filled with water. If a rectangular block of iron 14 cm in length, 10.5 cm in breadth and 11 cm in thickness is wholly immersed in water, by how many centimetres will the water level rise?

Solution Volume of the block $= 14 \times 10.5 \times 11 \text{ cm}^3$

Radius of the tank $= \frac{21}{2} = 10.5$ cm

Volume of the cylinder $= \pi r^2 h = \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times h$

$$\therefore \quad \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times h = 14 \times 10.5 \times 11$$

$$h = \frac{14}{3} = 4\frac{2}{3} \text{ cm}$$

Exp. 6) If the radius of cylinder is doubled, but height is reduced by 50%. What is the percentage change in volume?

Solution $\frac{r_1}{r_2} = \frac{r}{2r}$ and $\frac{h_1}{h_2} = \frac{h}{h/2}$

$\therefore$ Actual volume $= \pi r^2 h$

New volume $= \pi(2r)^2 \times \frac{h}{2} = 2\pi r^2 h$

Therefore new volume is the twice of the original volume.

Hence the change in volume $= \frac{2-1}{1} \times 100 = 100\%$

Exp. 7) The radius of the base of a right cone is 35 cm and its height is 84 cm. Find :

(a) slant height
(b) curved surface area
(c) total surface area
(d) volume

Solution (a) Slant height $(l) = \sqrt{r^2 + h^2}$

($r \rightarrow$ radius of the circular base)

$= \sqrt{35^2 + 84^2}$ ($h \rightarrow$ height of the cone)

$= \sqrt{1225 + 7056}$

$= \sqrt{8281} = 91$ cm

(b) Curved surface area $= \pi rl = \frac{22}{7} \times 35 \times 91$

$= 10010 \text{ cm}^2$

(c) Total surface area = lateral surface area + base area

$= \pi rl + \pi r^2 = \pi r\,(l + r)$

$= \frac{22}{7} \times 35\,(91 + 84)$

$= 110 \times 175 = 19250 \text{ cm}^2$

(d) Volume $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 35 \times 35 \times 84 = 107800 \text{ cm}^3$

Exp. 8) Find the area of the iron sheet required to prepare a cone 20 cm high with base radius 21 cm.

Solution $r = 21$ cm, $h = 20$ cm

$\therefore \quad l = \sqrt{r^2 + h^2} = 29$ cm

$\therefore$ Area of the sheet = total surface area of the cone

$= \pi rl + \pi r^2 = \pi r\,(l + r)$

$= \frac{22}{7} \times 21\,[29 + 21] = 3300 \text{ cm}^2$

Exp. 9) A solid metallic cylinder of base radius 3 cm and height 5 cm is melted to make n solid cones of height 1 cm and base radius 1 mm. Find the value of n.

Solution $n = \frac{\text{volume of cylinder}}{\text{volume of one cone}}$

$= \frac{\pi \times 3 \times 3 \times 5}{\frac{1}{3}\pi \times \frac{1}{10} \times \frac{1}{10} \times 1} = 13500$

Introductory *Exercise 10.6*

1. How many cubic metres of water fill a pipe which is 3500 m long and 0.08 m in diameter?
(a) 17.5 m^3 (b) 17.6 m^3 (c) 21 m^3 (d) 35 m^3

2. A cube of metal, whose edge is 10 cm, is wholly immersed in water contained in cylindrical tube whose diameter is 20 cm. By how much will the water level rise in the tube?
(a) 3.3 cm (b) $6\frac{3}{11}$ cm
(c) $3\frac{2}{11}$ cm (d) none of these

3. Find the height of the cylinder whose volume is 511 m^3 and the area of the base is 36.5 m^2 :
(a) 7 m (b) 10.5 m
(c) 14 m (d) none of these

4. The lateral surface area of a cylinder is 1056 cm^2 and its height is 16 cm. What is its volume?
(a) 5566 cm^3 (b) 4455 cm^3
(c) 5544 cm^3 (d) none of these

5. There is a cubical block of wood of side 2 cm. If the cylinder of the largest possible volume is curved out from it. Find the volume of the remaining wood :
(a) $\frac{7}{12}$ cm^2 (b) $\frac{12}{7}$ cm^3
(c) $5\frac{5}{7}$ cm^3 (d) none of these

6. The amount of concrete required to build a cylindrical pillar whose base has a perimeter of 8.8 m and whose curved surface area is 17.6 m^2 :
(a) 12.32 m^3 (b) 12.23 m^3
(c) 9.235 m^3 (d) 8.88 m^3

7. If the diameter of the base of a closed right circular cylinder be equal to its height 'h', then its whole surface area is :
(a) $\frac{2}{3}\pi h^2$ (b) $\frac{3}{2}\pi h^3$ (c) $\frac{3}{2}\pi h^2$ (d) πh^3

8. A right circular cylindrical tunnel of diameter 4 m and length 10 m is to be constructed from a sheet of iron. The area of the iron sheet required :
(a) $\frac{280}{\pi}$ (b) 40 π
(c) 80 π (d) none of these

9. The ratio between curved surface area and total surface area is 2 : 3. If the total surface area be 924 cm^2, find the volume of the cylinder :
(a) 2156 cm^3 (b) 1256 cm^3
(c) 1265 cm^3 (d) none of these

10. If the volume and curved surface area of a cylinder are 269.5 cm^3 and 154 cm^2 respectively, what is the height of the cylinder?
(a) 6 (b) 3.5
(c) 7 (d) can't be determined

11. If the curved surface area of a cylinder is 1320 cm^2 and its base radius is 21 cm, then its total surface area is :
(a) 4092 cm^2 (b) 2409 cm^2
(c) 4920 cm^2 (d) none of these

12. The ratio between the radius of the base and the height of a cylindrical pillar is 3 : 4. If its volume is 4851 m^3, the curved surface area of the pillar is :
(a) 924 m^2 (b) 1617 m^2
(c) 425 m^2 (d) none of these

13. If the ratio of total surface area to the curved surface area of a cylinder be 4 : 1, what is the ratio of radius to the height?
(a) 4 : 1 (b) 2 : 3
(c) 3 : 2 (d) 3 : 1

14. The circumference of the base of a right cylinder is 33 cm and height is 330 cm. What is the volume of this cylinder?
(a) 28586.25 cm^3 (b) 3344 cm^3
(c) 4433 cm^3 (d) 3456 cm^3

15. The radius of an iron rod decreased to one-fourth. If its volume remains constant, the length will become :
(a) 2 times (b) 8 times
(c) 4 times (d) 16 times

16. The total surface area of the cylinder is 2640 m^2 and the sum of height and radius of base of cylinder is 30 m. What is the ratio of height and radius of the cylinder?
(a) 7 : 9 (b) 9 : 7
(c) 8 : 7 (d) 3 : 7

17. The radii of two cylinders are in the ratio of 3 : 5 and their heights are in the ratio 4 : 3. The ratio of their volumes is :
(a) 12 : 25 (b) 13 : 25
(c) 4 : 5 (d) 5 : 4

18. The heights of two cylinders are in the ratio of 3 : 1. If the volumes of two cylinders be same, the ratio of their respective radii are :
(a) $\sqrt{3}$:1 (b) 1 : $\sqrt{3}$
(c) 1 : 9 (d) none of these

19. The ratio of heights of two cylinders is 3 : 2 and the ratio of their radii is 6 : 7. What is the ratio of their curved surface areas?
(a) 9 : 7 (b) 1 : 1
(c) 7 : 9 (d) 7 : 4

20. A hollow garden roller 42 cm wide with a girth of 132 cm is made of iron 3 cm thick. The volume of the iron of the roller is :
(a) 15544 cm^3
(b) 15444 cm^3
(c) 15545 cm^3
(d) none of the above

21. A conical vessel has a capacity of 15 L of milk. Its height is 50 cm and base radius is 25 cm. How much milk can be contained in a vessel in cylindrical form having the same dimensions as that of the cone?
(a) 15 L (b) 30 L
(c) 45 L (d) none of these

22. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $\frac{1}{27}$ of the volume of the given cone, at what height above the base is the section made?
(a) 20 cm (b) 18 cm (c) 27 cm (d) 15 cm

23. A tent is in the form of right circular cone 10.5 m high, the diameter of the base being 13 m. If 8 men are in the tent, find the average number of cubic metres of air space per man.
(a) $32\frac{3}{58}$ (b) 59.75 (c) $36\frac{9}{13}$ (d) $58\frac{3}{32}$

24. The radius and height of a right circular cone are in the ratio of 5 : 12. If its volume is $314\frac{2}{7}$ m^3, its slant height is :
(a) 26 m (b) 19.5 m
(c) 13 m (d) none of these

25. The volume and height of a right circular cone are 1232 cm^3 and 24 cm respectively, the area of its curved surface (in cm^2) is :
(a) 1100 (b) 225 (c) 616 (d) 550

26. The circumference of the base of a right circular cone is 220 cm^3 and its height 84 cm. The curved surface area of the cone is :
(a) 20020 cm^2 (b) 2020 cm^2
(c) 2200 cm^2 (d) 10010 cm^2

27. How many metres of cloth 10 m wide will be required to make a conical tent with base radius of 14 m and height is 48 m?
(a) 110 m (b) 55 m (c) 77 m (d) 220 m

28. A cone of height 2.8 cm has a lateral surface area 23.10 m^2. The radius of the base is :
(a) 3.5 cm (b) 2 cm (c) 2.1 cm (d) 4 cm

29. The radii of two cones are equal and their slant heights are in the ratio 3 : 2. If the curved surface area of the smaller cone is 300 cm^2, then the curved surface area of the bigger cone (in cm^2) is :
(a) 250 (b) 450 (c) 150 (d) 200

30. The ratio of the volume of a right circular cylinder and a right circular cone of the same base and height will be :
(a) 2 : 3 (b) 1 : 3 (c) 3 : 1 (d) 9 : 1

31. If the diameter of the base of right circular cone is equal to 8 cm and its slant height is 5 cm, then the area of its axial section is :
(a) 9 cm^2 (b) 12 cm^2 (c) 20 cm^2 (d) 40 cm^2

32. If the base radius and the height of a right circular cone are increased by 40%, then the percentage increase in volume (approx) is :
(a) 175% (b) 120%
(c) 64% (d) 540%

33. From a circular sheet of paper of radius 25 cm, a sector area 4% is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and height of the cone is :
(a) 16 : 25 (b) 9 : 25
(c) 7 : 12 (d) 24 : 7

34. If the radius of the base is doubled, keeping the height constant, what is the ratio of the volume of the larger cone to the smaller cone?
(a) 2 : 1 (b) 3 : 1 (c) 4 : 1 (d) 4 : 3

35. A largest possible cone is cut out from a cube of volume 1000 cm^3. The volume of the cone is :
(a) 280 cm^3 (b) 261.9 cm^3
(c) 269.1 cm^3 (d) 296.1 cm^3

36. If the height and the radius of a cone are doubled, the volume of the cone becomes :
(a) 2 times (b) 8 times
(c) 16 times (d) 4 times

37. A conical tent has 60° angle at the vertex. The ratio of its radius and slant height is :
(a) 3 : 2 (b) 1 : 2
(c) 1 : 3 (d) can't be determined

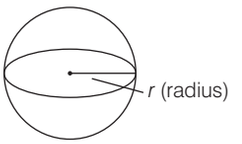
38. Water flows at the rate of 5 m per min from a cylindrical pipe 16 mm in diameter. How long will it take to fill up a conical vessel whose radius is 12 cm and depth is 35 cm?
(a) 315 s (b) 365 s
(c) 5 min (d) none of these

39. A reservoir is in the shape of a frustum of a right circular cone. It is 8 m across at the top and 4 m across at the bottom. It is 6 m deep its capacity is :
(a) 224 m^3 (b) 176 m^3
(c) 225 m^3 (d) none of these

40. A conical vessel whose internal radius is 10 cm and height 72 cm is full of water. If this water is poured into a cylindrical vessel with internal radius 30 cm, the height of the water level rises in it is :
(a) $2\frac{2}{3}$ cm (b) $3\frac{2}{3}$ cm
(c) $5\frac{2}{3}$ cm (d) none of these

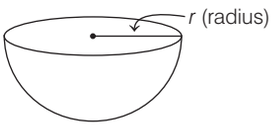
10.10 Sphere, Prism and Pyramid

Sphere



$$\text{Volume} = \frac{4}{3}\pi r^3, \quad \text{Surface area} = 4\pi r^2$$

Hemisphere :



$$\text{Volume} = \frac{2}{3}\pi r^3$$

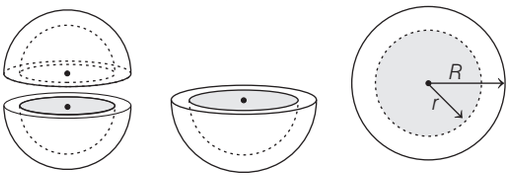
$$\text{Curved surface area} = 2\pi r^2$$

$$\text{Total surface area} = 3\pi r^2 = (2\pi r^2 + \pi r^2)$$

Spherical shell :

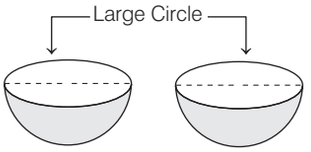
$$\text{Volume} = \frac{4}{3}\pi\,(R^3 - r^3)$$

$$\text{Total surface area} = 4\pi\,(R^2 + r^2)$$



Circles in the Sphere

(i) **Great Circle:** If a plane cuts the sphere at the centre, we get two hemispheres of equal volume and equal surface area. In this case, the circles that we get are called great circles, since these are the circles with the greatest possible chords. The longest chord (or diameter) of the great circle is same as the longest chord (or diameter) of the sphere.



(ii) **Small Circle:** If a plane cuts the spheres in such a way that the cut is not made through the centre of the sphere, then we get two frustums of unequal volume and unequal lateral surface area, but the area of the two circles would be same. However, these circles would be called as small circles, since the diameter of these circles is shorter than the diameter of the sphere.

